IGCSE 0607 Practice Questions — E8.1

Transformations

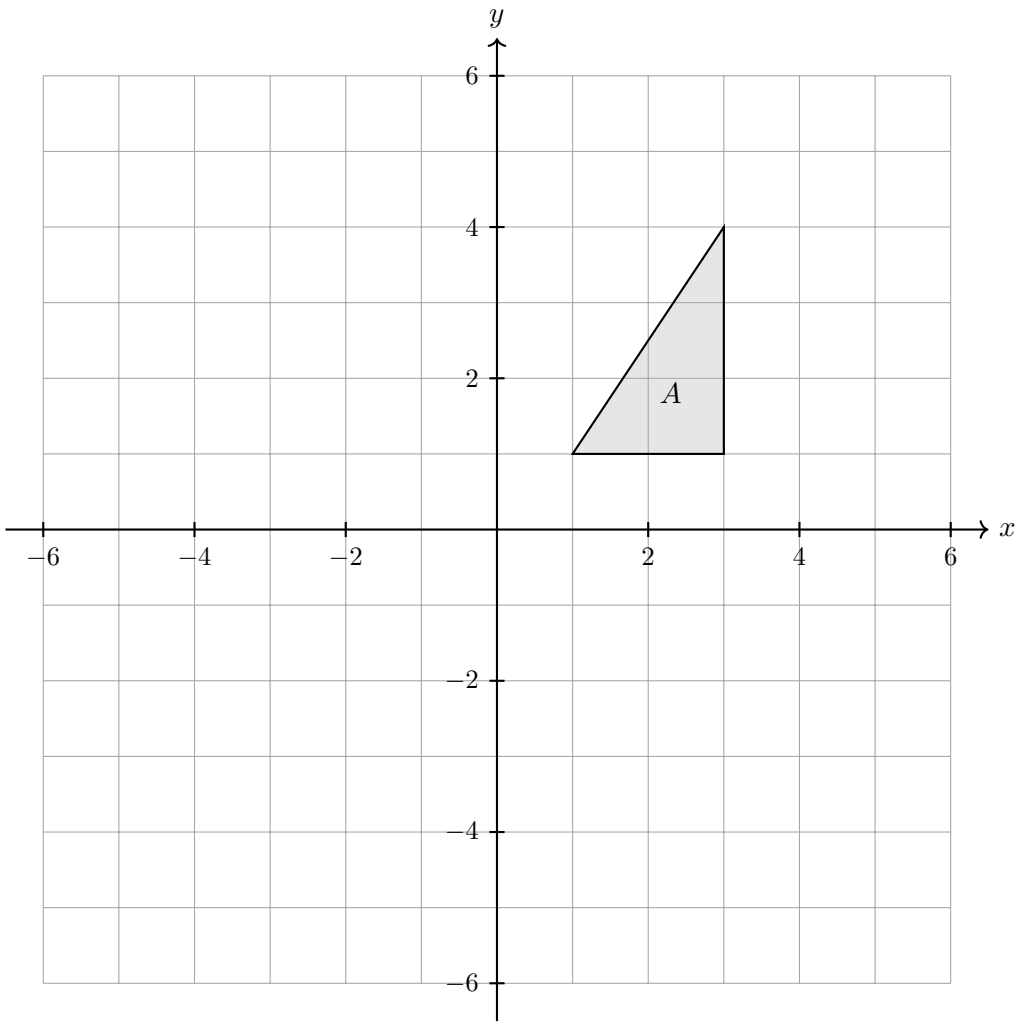
Total Questions: 24 — **Total Marks:** 158 Marks

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- **E8.1** Transformations
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

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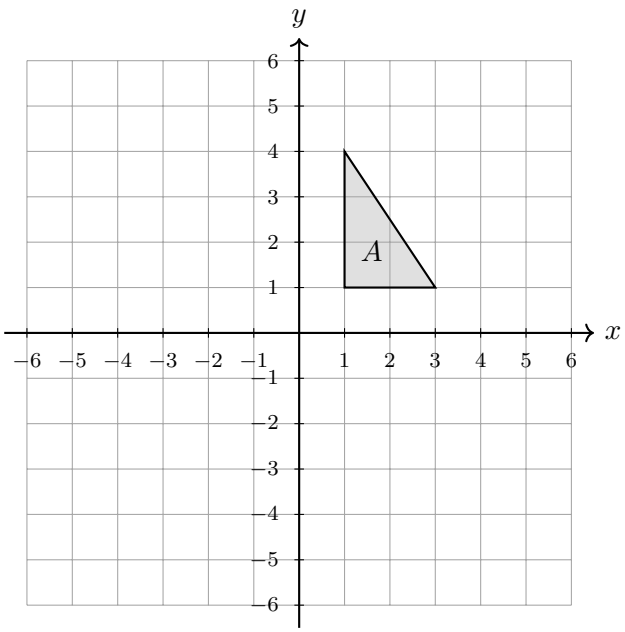
1 The grid shows triangle A.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $x = -1$.
Label the image C. **[2]**

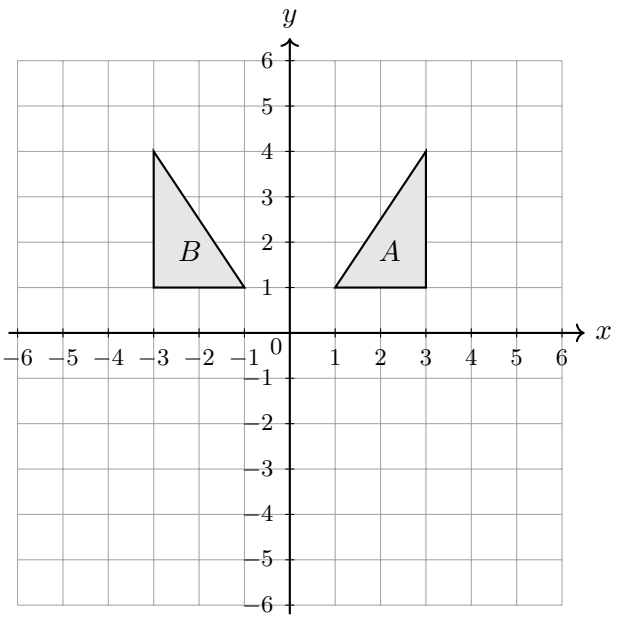
2 The diagram shows a triangle, Shape A, drawn on a coordinate grid.



(a) On the grid, draw the image of Shape A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label this image B. **[2]**

(b) On the grid, draw the image of Shape A after a reflection in the line $y = -1$.
Label this image C. **[2]**

3 The diagram shows two triangles, A and B, plotted on a coordinate grid.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$. **[2]**

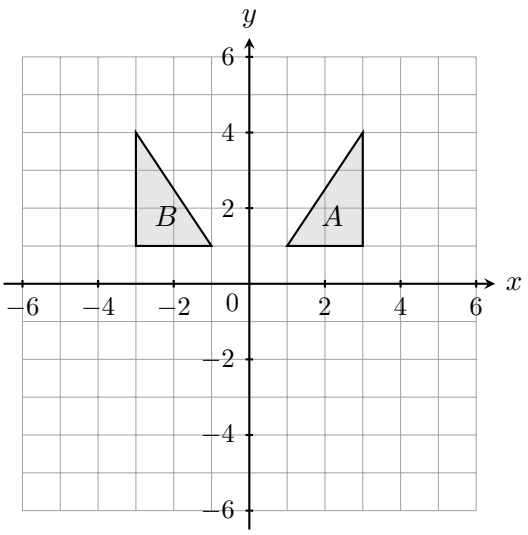
(b) Describe fully the single transformation that maps triangle A onto triangle B.

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..... **[2]**

4 The grid shows triangle A and triangle B.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$. **[2]**

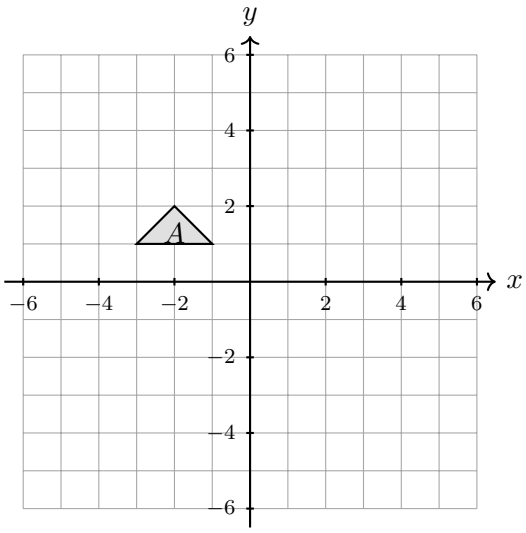
(b) Describe fully the single transformation that maps triangle A onto triangle B.

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[2]

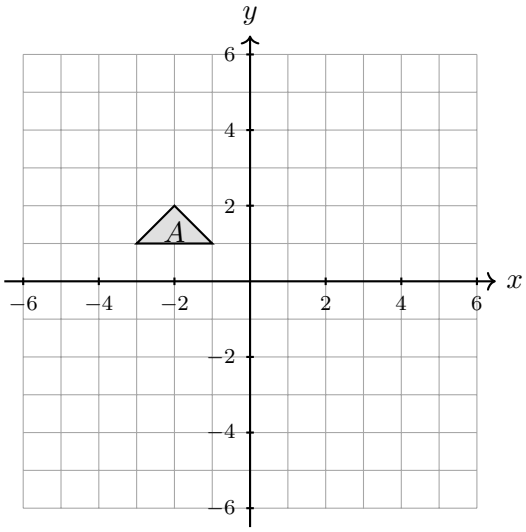
5 A triangle A is drawn on a grid.

(a) On the grid above, draw the image of triangle A after a translation by the vector $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.



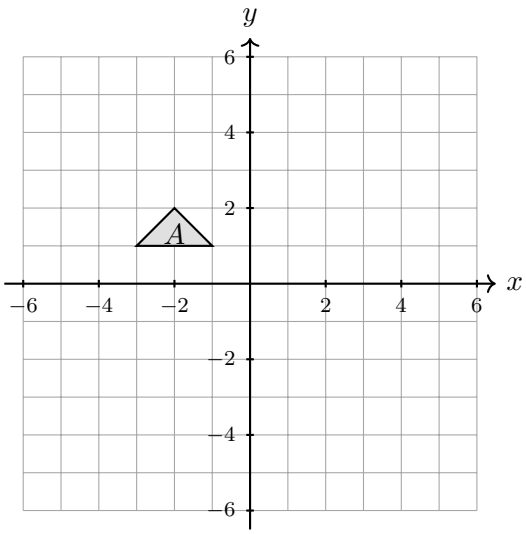
[2]

(b) On the grid above, draw the image of triangle A after a reflection in the line $y = x$.



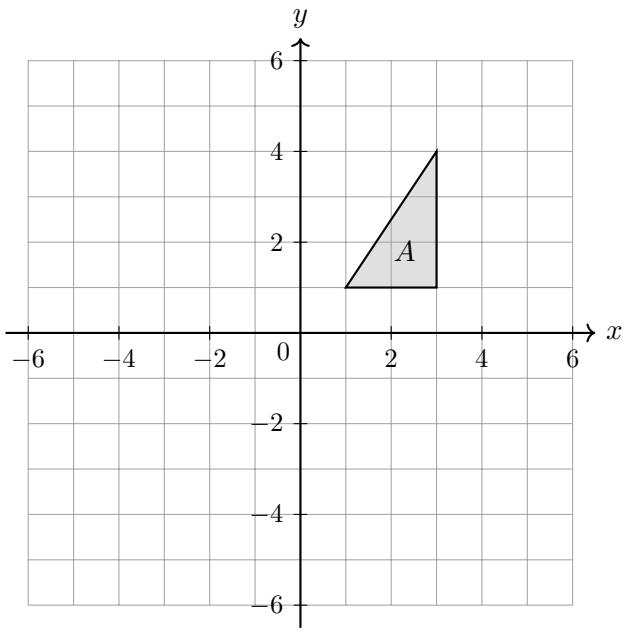
[2]

(c) On the grid above, draw the image of triangle A after an enlargement with scale factor 2 and centre $(-1, 0)$.



[2]

6 The diagram shows triangle A on a coordinate grid.

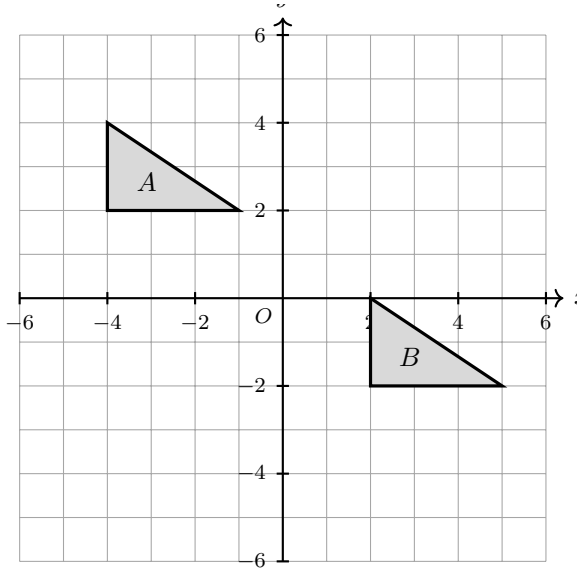


(a) On the grid, translate shape A by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, reflect shape A in the x-axis.
Label the image C. **[2]**

(c) On the grid, rotate shape A by 180° about the origin $(0, 0)$.
Label the image D. **[2]**

7 The diagram shows triangles A and B on a coordinate grid.



(a) Describe fully the single transformation that maps triangle A onto triangle B.

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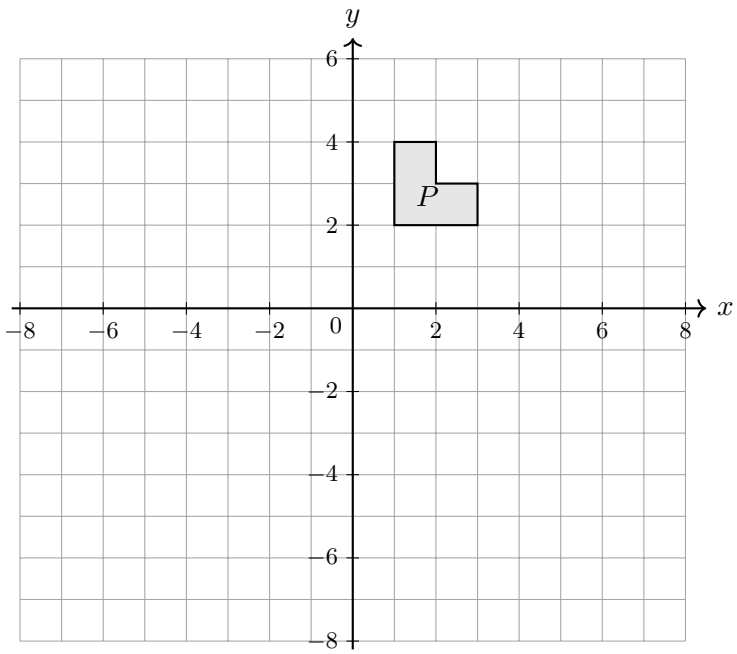
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..... **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $y = 1$.
Label the image C. **[2]**

(c) On the grid, draw the image of triangle A after a rotation of 90° clockwise about the origin $(0, 0)$.
Label the image D. **[3]**

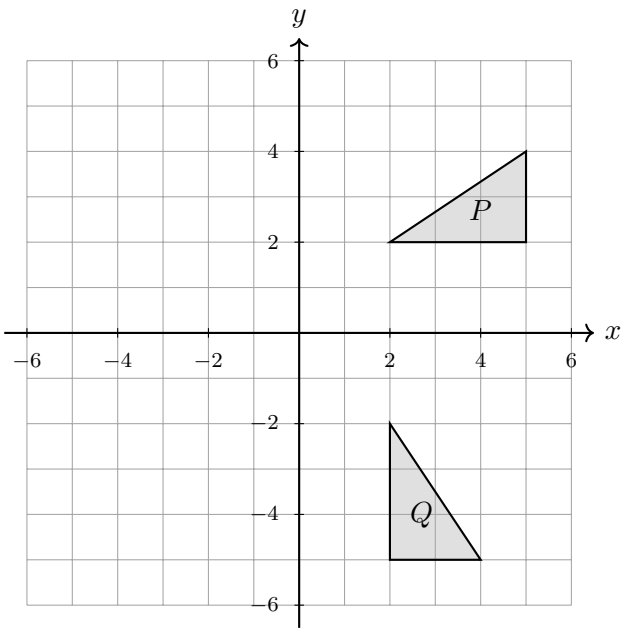
8 The diagram shows a shape P plotted on a coordinate grid.



(a) On the grid, draw the image of shape P after an enlargement, scale factor -2, centre $(0, 0)$. **[2]**

(b) On the grid, draw the image of shape P after a rotation of $90°$ clockwise about the point $(1, -1)$. **[2]**

9 The diagram shows two triangles, Shape P and Shape Q, drawn on a grid.



(a) Describe fully the single transformation that maps Shape P onto Shape Q.

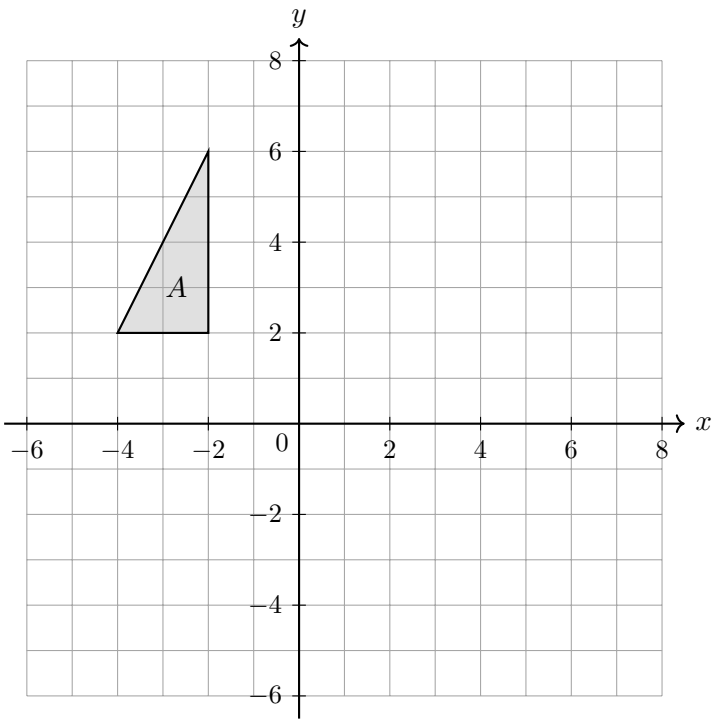
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..... **[3]**

(b) Shape P is enlarged by a scale factor of $\frac{1}{2}$ with centre of enlargement $(-2, 2)$.
On the grid, draw the image of Shape P and label it R. **[2]**

10 The diagram shows shape A on a coordinate grid.



(a) Shape A is reflected in the line $y = x$ to give shape B.
On the grid, draw and label shape B. **[2]**

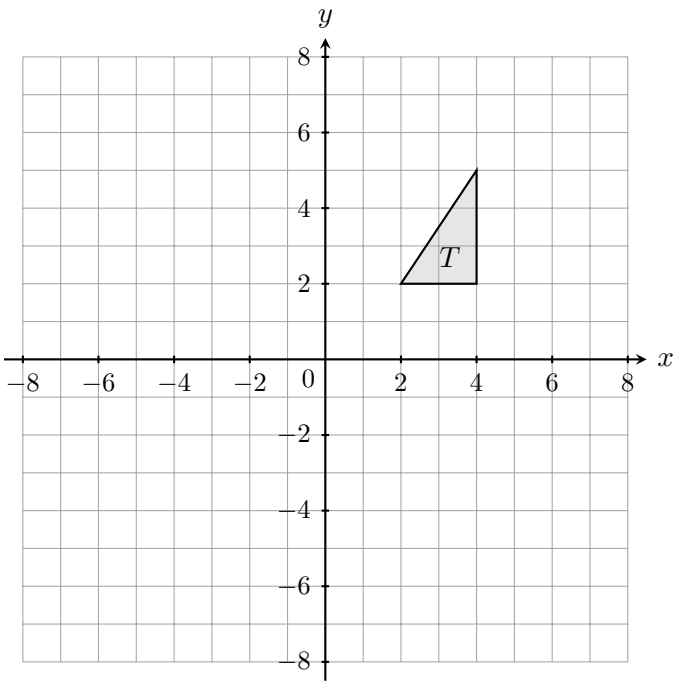
(b) Shape B is rotated by 90° clockwise about the point $(2, 2)$ to give shape C.
On the grid, draw and label shape C. **[2]**

(c) Describe fully the single transformation that maps shape A onto shape C.

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..... **[2]**

11 The grid shows triangle T.



(a) On the grid, draw the image of triangle T after an enlargement with scale factor $-\frac{1}{2}$ and centre $(0, 2)$. **[2]**

(b) Triangle T is reflected in the line $x = 1$ to give triangle U.
Triangle U is then reflected in the line $y = 2$ to give triangle V.
Describe fully the single transformation that maps triangle T onto triangle V.

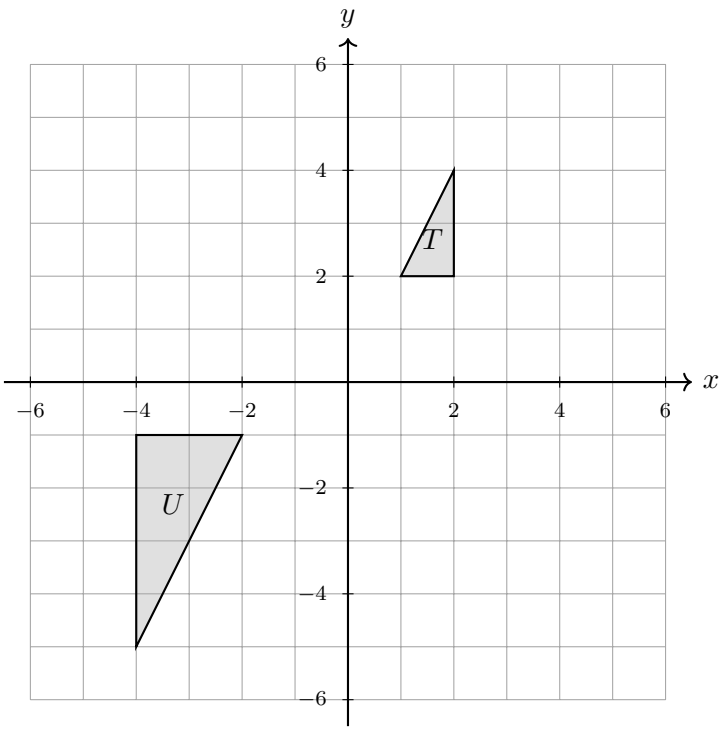
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[3]

(c) Triangle T is enlarged by a scale factor of k to give triangle W.
The area of triangle W is 48 square units.
Find the value of k, given that $k > 0$.

$k =$ ..... **[2]**

12 The diagram shows two triangles, Shape T and Shape U.



(a) Describe fully the single transformation that maps Shape T onto Shape U.

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..... **[3]**

(b) Describe fully the single transformation that maps Shape U onto Shape T.

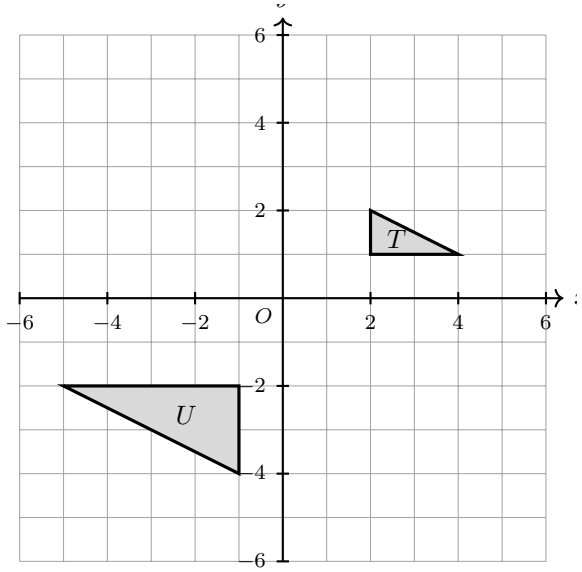
.....

..... **[2]**

(c) Shape T is translated by the vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ to give Shape V.
Shape V is then reflected in the line $y = x$ to give Shape W.
Find the coordinates of the three vertices of Shape W.

..... **[3]**

13 The diagram shows triangles T and U on a coordinate grid.



(a) Describe fully the single transformation that maps triangle T onto triangle U.

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..... **[3]**

(b) Triangle T is reflected in the line $x = 1$ to give triangle V.
Triangle V is then reflected in the line $y = 2$ to give triangle W.

(i) On the grid, draw and label triangle W. **[3]**

(ii) Describe fully the single transformation that maps triangle T directly onto triangle W.

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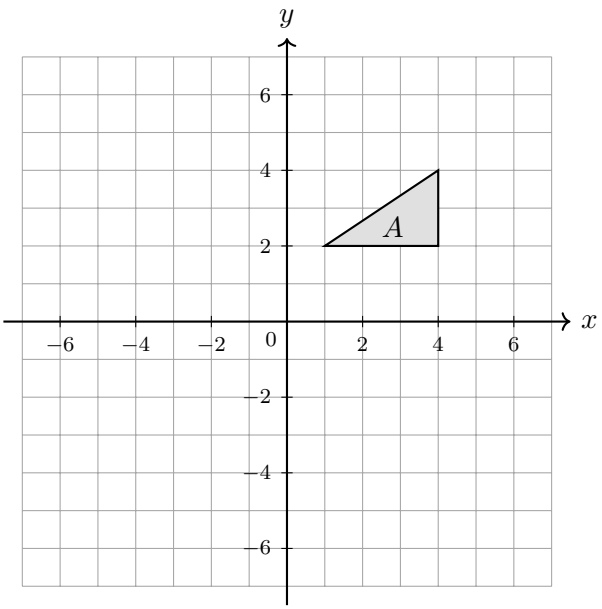
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[3]

(c) Point $P(a, b)$ is mapped onto the point $P'(5, -3)$ by a reflection in the line $y = x$ followed by a translation by the vector $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$.
Find the coordinates of P.

$P = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[3]**

14 The diagram shows triangle A on a coordinate grid.

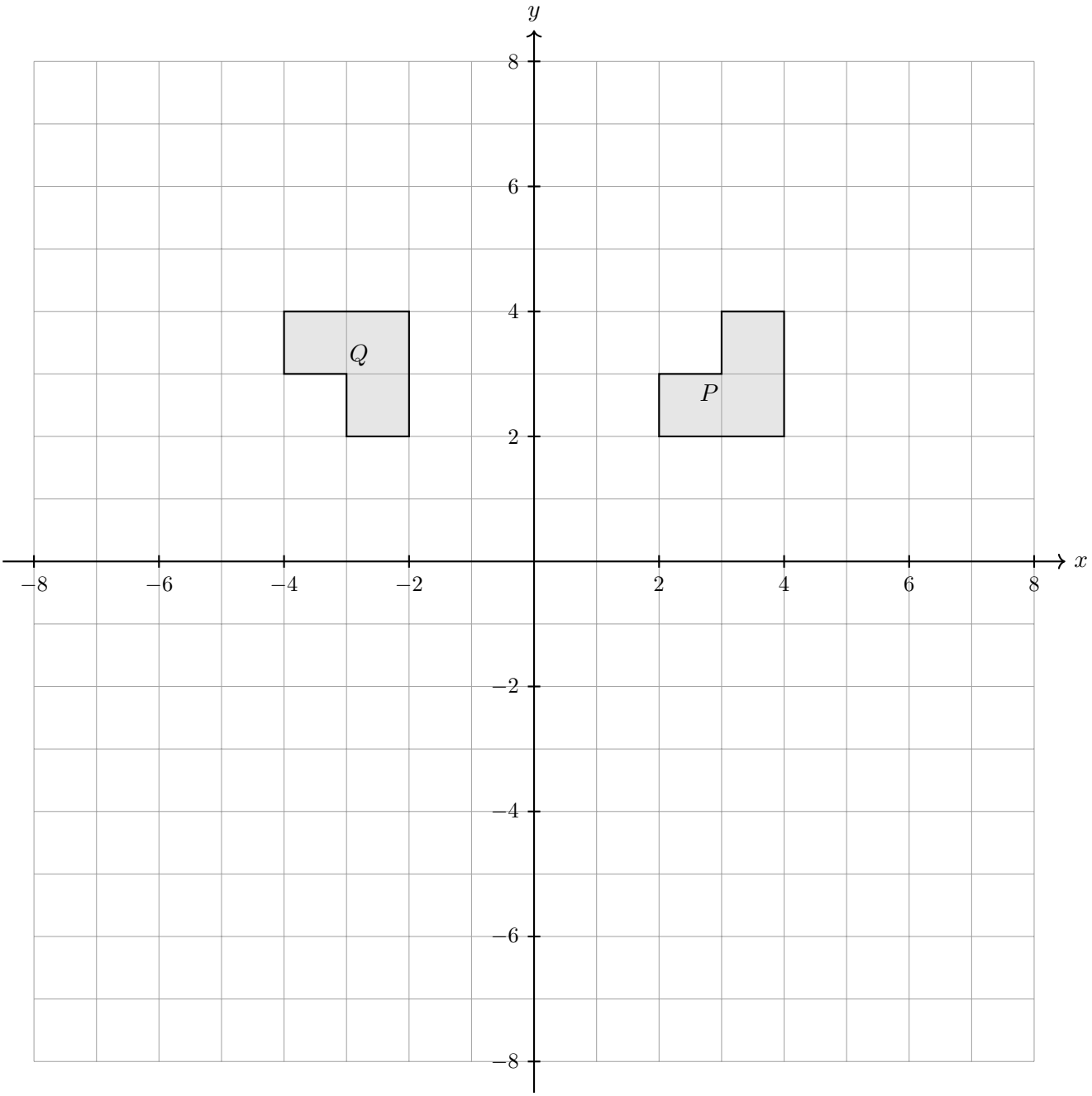


(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -6 \\ -3 \end{pmatrix}$. **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $y = 1$. **[2]**

(c) On the grid, draw the image of triangle A after a rotation of $180°$ about the origin $(0, 0)$. **[2]**

15 The grid shows shapes P and Q.



(a) Describe fully the single transformation that maps shape P onto shape Q.

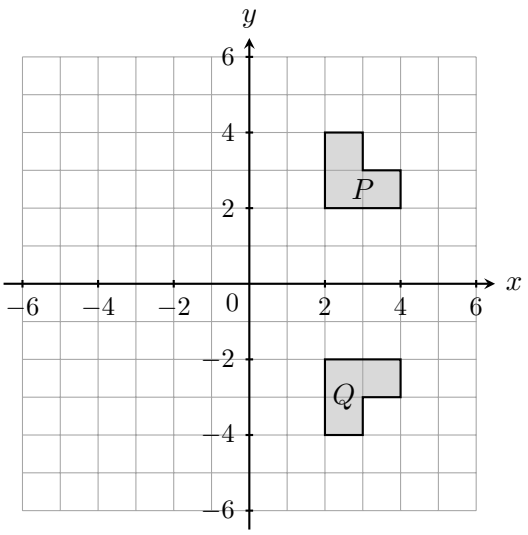
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..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$, centre $(6, 6)$.
Label the image R. **[2]**

16 The grid shows shapes P and Q.



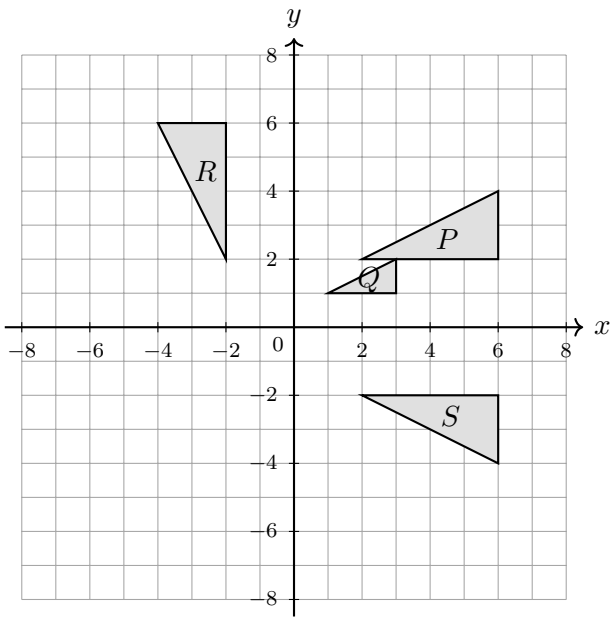
(a) On the grid, draw the image of shape P after an enlargement with scale factor 0.5 and centre $(0, 0)$. **[2]**

(b) Describe fully the single transformation that maps shape P onto shape Q.

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[3]

17 The diagram shows triangles P, Q, R, and S on a coordinate grid.



(a) Describe fully the single transformation that maps triangle P onto triangle Q.

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..... **[3]**

(b) Describe fully the single transformation that maps triangle P onto triangle R.

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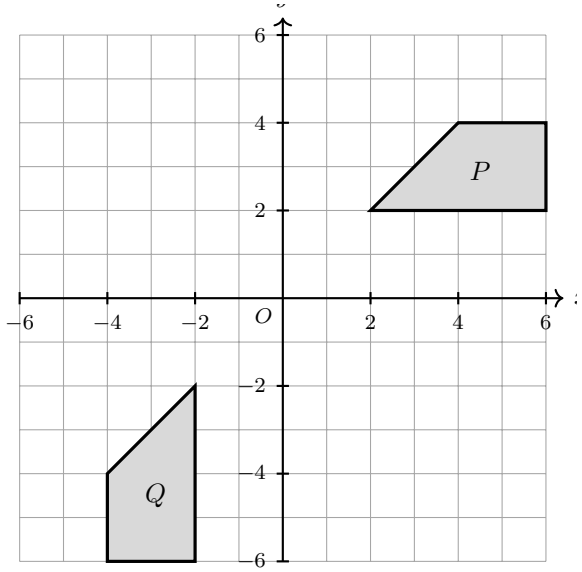
..... **[3]**

(c) Describe fully the single transformation that maps triangle P onto triangle S.

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..... **[2]**

18 The diagram shows shapes P and Q on a coordinate grid.



(a) Describe fully the single transformation that maps shape P onto shape Q.

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..... **[2]**

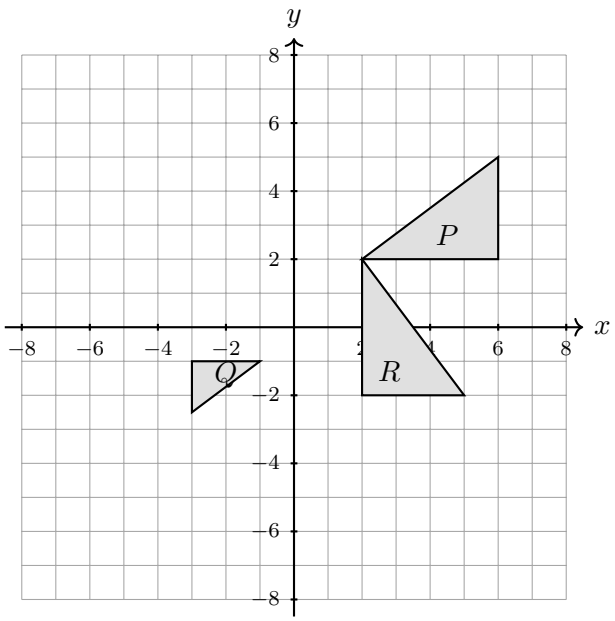
(b) On the grid, draw the image of shape P after an enlargement with scale factor $\frac{1}{2}$ and centre $(-2, -2)$.
Label the image R. **[3]**

(c) On the grid, draw the image of shape P after a rotation of 90° anticlockwise about the point $(1, 1)$.
Label the image S. **[3]**

(d) Find the ratio of the area of shape R to the area of shape P, giving your answer in its simplest form.

..... **[1]**

19 The diagram shows triangles P, Q and R.



(a) Describe fully the single transformation that maps triangle P onto triangle Q.

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..... **[3]**

(b) Describe fully the single transformation that maps triangle P onto triangle R.

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..... **[3]**

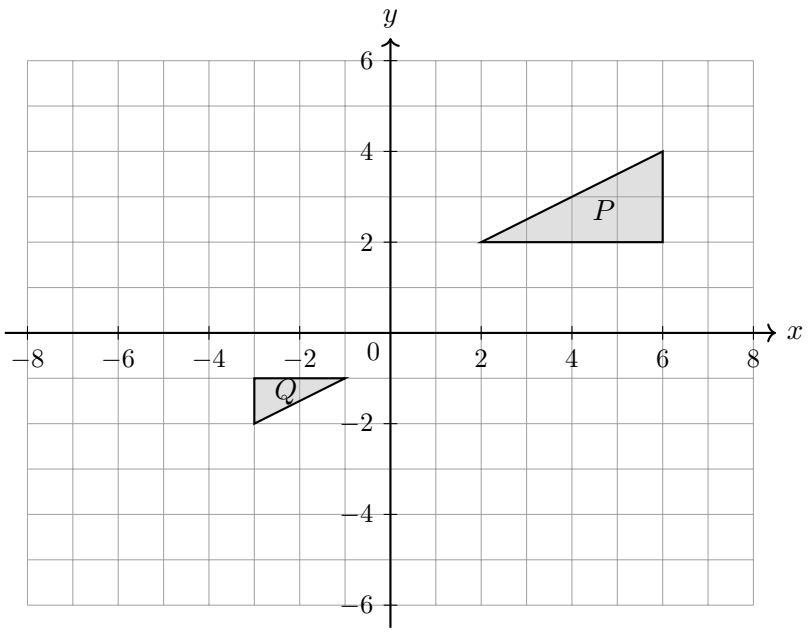
(c) Triangle Q is reflected in the origin $(0, 0)$ to give triangle S.
Describe fully the single transformation that maps triangle P onto triangle S.

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..... **[3]**

20 The diagram shows triangles P and Q on a coordinate grid.



(a) Describe fully the single transformation that maps shape P onto shape Q.

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..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$ with centre $(0, 4)$.
Label the image R. **[3]**

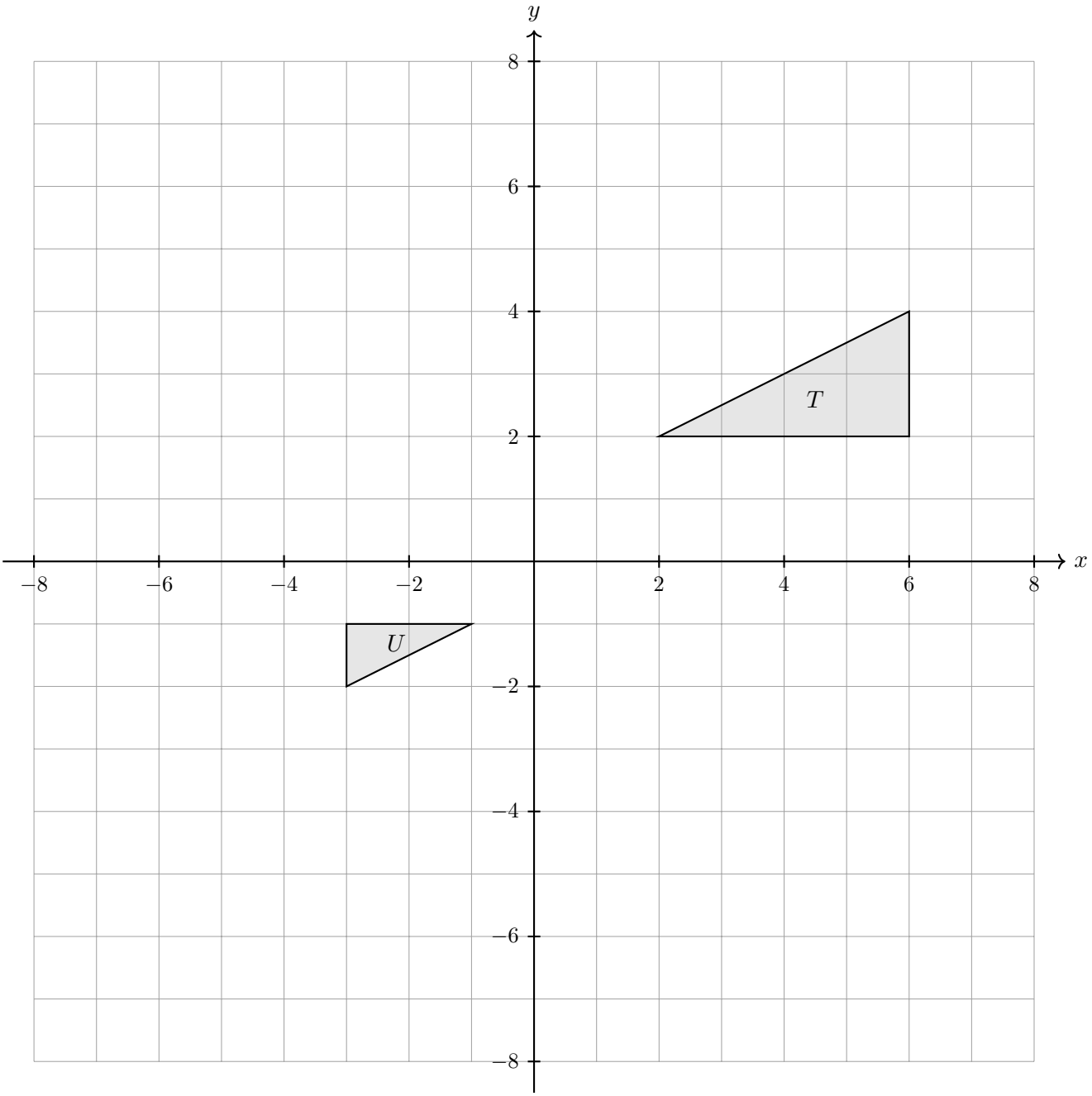
(c) Describe fully the single transformation that maps shape R onto shape Q.

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..... **[3]**

21 The grid shows triangles T and U.



(a) Describe fully the single transformation that maps triangle T onto triangle U.

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..... **[3]**

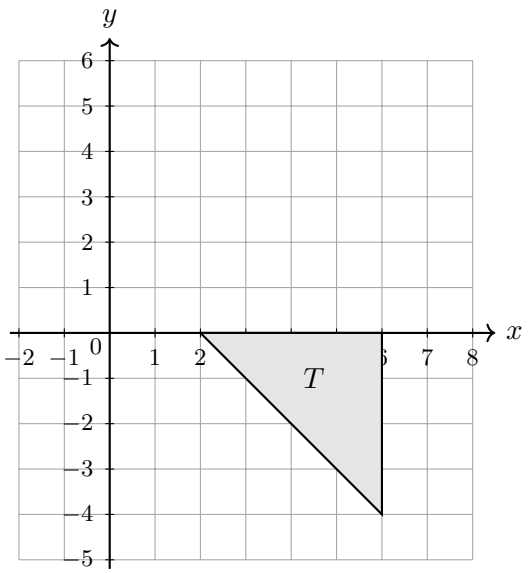
(b) Triangle T is reflected in the line $y = x$ to give triangle V.
Triangle V is then rotated 90° clockwise about the origin to give triangle W.
Describe fully the single transformation that maps triangle T onto triangle W.

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..... **[2]**

22 The diagram shows triangle T plotted on a coordinate grid.



(a) Triangle T is mapped onto triangle U by an enlargement, scale factor $\frac{1}{2}$, centre $(0, 4)$.
On the grid, draw triangle U. **[2]**

(b) Triangle U is mapped onto triangle V by a reflection in the line $y = x$.
On the grid, draw triangle V. **[2]**

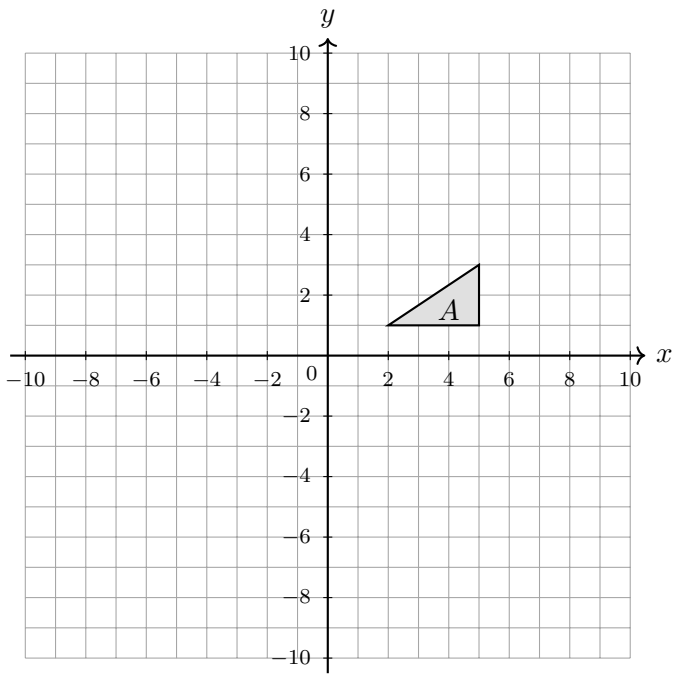
(c) Describe fully the single transformation that maps triangle U back onto triangle T.

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..... **[3]**

23 The diagram shows triangle A on a coordinate grid.



(a) On the grid, draw the image of triangle A after an enlargement by scale factor -2, centre $(1, 0)$. Label this image B. **[3]**

(b) Triangle A is mapped onto triangle C by a reflection in the line $y = -x$.
Write down the coordinates of the vertices of triangle C.

..... **[2]**

(c) Triangle A is first translated by the vector $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$.

The resulting image is then rotated 90° clockwise about the point $(5, 5)$ to form triangle D.
Describe fully the single transformation that maps triangle A onto triangle D.

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..... **[3]**

24 A triangle T has vertices at (a, b), (c, d) and (e, f).
The translation by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ maps triangle T onto triangle U with vertices at $(5, 2)$, $(8, 2)$ and $(7, 6)$.

(a) Find the coordinates of the three vertices of triangle T.

..... **[2]**

(b) Triangle U is mapped onto triangle V by a reflection in the line $y = x$.
Write down the coordinates of the three vertices of triangle V.

..... **[2]**

(c) Triangle V is mapped onto triangle W by a rotation of $90°$ clockwise about the point $(2, 5)$.
Find the coordinates of the three vertices of triangle W.

..... **[3]**

(d) Describe fully the single transformation that maps triangle T onto triangle X, where the vertices of X are $(-4, -8)$, $(-10, -8)$ and $(-8, -16)$.

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..... **[3]**